

Solid equilibria and adsorption

After midterm · Session 4 of 6 · 180 minutes

Soorathep Kheawhom

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The question for today

What happens to gas pressure when the solid inventory runs out?

Before class: 3-page reading · 15–20 minutes

Bring the three preparation answers. Revisit your prediction after one controlled comparison and an independent check.

The 180-minute class

ACTIVITY	MINUTES
Opening question and essential concepts	30
One hand calculation	20
One lab demonstration	20
Break	10
Student pair investigation	70
Discussion and exit question	30

The 70-minute investigation is student work with instructor support.

Name the physical process first

PROCESS	WHAT CHANGES?
SLE / sublimation	The same substance transfers between bulk phases
Adsorption	Surface uptake changes
Gas–solid reaction	Chemical species change through a reaction

Lab 10 covers nonreacting bulk equilibria; Lab 11 covers adsorption. General heterogeneous reaction solving is outside these labs.

Sublimation and vessel inventory

At fixed T , a present bulk solid fixes the ideal-gas partial pressure at its sublimation pressure within the model.

In a rigid vessel,

$$n_g^{sat} = \frac{P_{sub}(T)V}{RT}$$

If total subliming-species inventory is below this amount, the solid is exhausted and pressure falls below P_{sub} .

A gas–solid inventory check

Synthetic example: $T=300\text{ K}$, $V=1\text{ L}$, $P_{\text{sub}}=10\text{ kPa}$.

$$n_g^{\text{sat}} = 10 / (8.31446 \times 300) \approx 0.00401\text{ mol}$$

With 0.01 mol total, about 0.00599 mol remains solid. With 0.002 mol total, no solid remains and $P \approx 4.99\text{ kPa}$.

Fixed T, V equilibrium minimizes Helmholtz energy, not Gibbs energy at fixed P .

An independent check

Compare total inventory with $P_{\text{sub}} V/(RT)$. Check gas plus solid moles and distinguish partial pressure from total pressure if an inert gas is present.

One demonstration • 20 minutes

Open the core lab

In Lab 10 choose the gas–solid vessel mode. Keep T and V fixed and vary the subliming-species inventory across the solid-depletion threshold.

Keep one saved baseline for the pair investigation.

Pair investigation • 70 minutes

- 10 min: predict and record the baseline.
- 25 min: change one input and calculate.
- 20 min: exchange roles and check the result by hand.
- 15 min: prepare one plot or table and a short explanation.

In Lab 10, compare inventories above and below the solid-depletion boundary at fixed T and V . Check the inventory balance and explain the resulting gas pressure.

Discussion and exit question • 30 minutes

Use 20 minutes to compare results and discuss unexpected behavior.

What happens to gas pressure when the solid inventory runs out?

Use the final 10 minutes to explain your answer individually and name one assumption that limits it.

Follow-up practice • 30–40 minutes

In Lab 10, compare inventories above and below the solid-depletion boundary at fixed T and V . Check the inventory balance and explain the resulting gas pressure.

Assessment 4 and independent checkpoint

Retain one study JSON, one comparison and a short explanation. Continue from your classroom case. The hand checkpoint is included. With pre-class reading, independent work totals 45–60 minutes.

Optional study

Choose one extension: compare Henry and Langmuir in Lab 11, check a closed adsorption vessel, or explore pure-solid solubility and the eutectic assumptions in Lab 10.

Optional paper exercises: Henry gas solubility and dilute-solution osmosis.

[Session appendix](#) · [Module reference](#)

[Learning path](#) · [Offline package](#)

Optional work is additional. Complete the required investigation first.